

# Draft IEEE Article - NetGuard AI

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## Abstract

Network reliability and security are critical for educational infrastructure. NetGuard AI proposes a lightweight machine learning anomaly detection system using CICIDS2017-style data, classical ML baselines, Random Forest as the proposed model, and a Flask dashboard with risk classification and recommended actions.

## Introduction

The project addresses a practical gap: many IDS studies stop at model metrics, while beginner administrators need dashboard interpretation and operational guidance.

## Method

The workflow includes preprocessing, binary label conversion, train/test split, Logistic Regression, Decision Tree, Random Forest, model selection, prediction, and dashboard visualization.

## Results

Current sample results are smoke-test artifacts only. Final submission must use a real CICIDS2017 subset and replace all values with real metrics.

## Conclusion

NetGuard AI is a reproducible educational prototype and can be improved with cross-validation, hyperparameter tuning, feature importance, and live network ingestion.

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